Table 19.1 Differences between Tropic and Nastic movements

Tropic movements	Nastic movements
Unidirectional response to the stimulus.	Non-directional response to the stimulus.
Growth dependent movements.	Growth independent movements.
More or less permanent and irreversible.	Temporary and reversible.
Found in all plants.	Found only in a few specialized plants.
Slow action.	Immediate action.

Thermonasty: Movement of part of a plant is associated with change in temperature. e.g. *Tulip* flowers bloom as the temperature increases.



Figure 19.5 Thermonasty in Tulip

19.3 Photosynthesis

'Photo' means 'light' and 'synthesis' means 'to build'. Thus photosynthesis literally means 'building up with the help of light'. During this process, the light energy is converted into chemical energy. Green plants are autotrophic in their mode of nutrition because they prepare their food materials through a process called photosynthesis. The overall equation of photosynthesis can be given as below:

$$\underset{\text{Carbon dioxide}}{6CO_2} + \underset{\text{Water}}{12H_2O} \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} \underset{\text{Gluose}}{C_6H_{12}O_6} + \underset{\text{Water}}{6H_2O} + \underset{\text{Oxygen}}{6O_2}\uparrow$$



Do the insects also trap solar energy? Tel Aviv University Scientists have found out that *Vespa orientalis* (Oriental Hornets) have similar capabilities to trap solar energy. They have a yellow patch on its abdomen and an unusual cuticle structure which is a stack of 30 layers thick. The cuticle does not contain chlorophyll but it contains the yellow light sensitive pigment called xanthopterin. This works as a light harvesting molecule transforming light energy into electrical energy.

The end product of photosynthesis is glucose which will be converted into starch and stored in the plant body. Plants take in carbon dioxide for photosynthesis; but for its living, plants also need oxygen to carry on cellular respiration.

19.3.1 Requirements for Photosynthesis'

Activity 3

Pluck a variegated leaf from *Coleus* plant kept in sunlight. De-starch it by keeping in dark room for 24 hours. Draw the picture of this leaf and mark the patches of cholorphyll on the leaf. Immerse the leaf in boiling water followed by alcohol and test it for starch using iodine solution.Record your observation.



Activity 4



Place a potted plant in a dark room for about 2 days to de-starch its leaves. Cover one of its leaves with the thin strip of black paper as shown in the picture. make sure that the leaf is covered on both sides. Keep the potted plant in bright sunlight for 4 to 6 hours. Pluck the selected covered leaf and remove the black paper. Immerse the leaf in boiling water for a few minutes and then in alcohol to remove chlorophyll. Test the leaf now with iodine solution for the presence of starch. The covered part of the leaf does not turn blue-black whereas the uncovered part of the leaf turns blue-black colour.

Why are the changes in colour noted in the covered and uncovered part of the leaf?

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